

1. You are buying food from the BRACU canteen. Each food item costs 50 Taka. After each order, ask the user if they want to order again (1 for yes, 0 for no). Use a do-while loop to keep taking orders until the user enters 0. If the total cost becomes more than 300 Taka, print “Bhai, card e taka sesh!” and stop the ordering using break.
2. You drink one cup of coffee every hour while studying. Each cup increases your energy by 15 points. If you skip coffee (user enters 0), print “Feeling sleepy... ” and use continue to skip that hour. Use a for loop that runs for 8 hours. If energy exceeds 100 points, print “Too much caffeine! Stop now” and stop using break.
3. You are saving money to buy a new phone. Each day, you can save between 0 and 1000 Taka (entered by the user). Use a while loop to keep adding the amount until your total savings reach 20,000 Taka. If you save more than 800 Taka on a day, print “Great saving today!”. If your total savings exceed 25,000, print “Enough savings! Time to buy the phone” and stop using break.
4. Write a program that calculates the electricity bill for 6 months. For each month, input the number of units used. Each unit costs 8 Taka. If units exceed 300, apply a 10% fine for that month. Use a for loop to calculate and display the bill for each month and print the total bill at the end.
5. Each gym visit adds 10 points to your fitness score. If you miss a day (user enters 0), subtract 5 points. Use a do-while loop to update your score each day. Stop when the user enters -1. At the end, print your total fitness score and total gym visits.
6. A student attends classes until the total learning time reaches 3 hours. Each class session lasts 30 minutes. After every class, ask if the student was attentive (1 for yes, 0 for no). If not attentive, reduce 10 minutes from total effective learning time. Use a while loop to track progress and print the total effective time at the end.
7. There are 10 students in the lunch queue. If a student doesn't have money (user enters 0), print “Skipped (no balance)!” and use continue. Otherwise, print “Served successfully!”. At the end, print how many students were served.
8. You recharge your phone repeatedly with amounts entered by the user. Each recharge adds to your balance. If balance exceeds 500 Taka, print “Enough balance! Stop scrolling TikTok” and use break. Use a do-while loop to continue until the user stops.
9. Each day, a student uses X liters of water (input by user). The cost is 0.5 Taka per liter. Use a while loop to calculate total cost for 30 days. If usage in a day exceeds 100 liters, print “High usage! Please save water”. At the end, display the total monthly water bill.
10. Write a program that simulates your semester GPA tracking. Each course has 3 credit hours. Ask the user for marks of 5 courses. Convert marks to grade points (A = 4.0, B = 3.0, C = 2.0, etc.) using if-else statements. Use a for loop to calculate and print the total GPA. If GPA < 2.5, print “Try harder next semester!”

11. Write a C program to calculate the sum of the first n terms of a series, where: If the number is even, the term is: $\text{number} / (\text{number} + 2)$ If the number is odd, the term is: $1 / (\text{number} + 1)$ The user will input a positive integer n . The program will print the total sum of the series up to n terms, rounded to 6 decimal places.
12. Write a C program to calculate the product of the first n terms of a series: If the term number is even, the term is: $1 + (1 / \text{number})$ If the term number is odd, the term is: $1 - (1 / \text{number})$ Take n as input and display the product of all terms.
13. Write a C program that calculates the alternating sum of the first n natural numbers: Add odd numbers and subtract even numbers. Example: $1 - 2 + 3 - 4 + 5 - 6 + \dots$ up to n .
14. Write a C program to find the sum of the following series up to n terms: $\text{term}(i) = i / (i + 1)$ if i is odd, else $\text{term}(i) = i / (i + 2)$ Take n as input and print the result rounded to 6 decimal places.
15. Write a C program to calculate the harmonic series up to n terms, but skip every third term. The formula is: $\text{Sum} = 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n$, skipping terms where $i \% 3 == 0$.
16. Write a C program that finds the sum of squares of even numbers and cubes of odd numbers up to n . If number is even, $\text{term} = \text{number}^2$; if number is odd, $\text{term} = \text{number}^3$. Print the total sum.
17. Write a C program that calculates the sum of the first n terms of a series: If the term index is divisible by 3, $\text{term} = \text{index} / 2.0$ Else $\text{term} = 1 / (\text{index} + 1.0)$ Print the total sum rounded to 6 decimal places.
18. Write a C program to calculate the difference between the sum of fractions of even numbers and the sum of fractions of odd numbers up to n . For even numbers: add $1/\text{number}$; for odd numbers: subtract $1/\text{number}$.
19. Write a C program to find the sum of the series: $1/2 + 2/3 + 3/4 + 4/5 + \dots + n/(n + 1)$. Take n as input and print the sum with 6 decimal places.
20. Write a C program that sums the reciprocal of factorial values up to n : $\text{Sum} = 1/1! + 1/2! + 1/3! + \dots + 1/n!$ Use loops to calculate each factorial term and display the final result rounded to 6 decimal places.